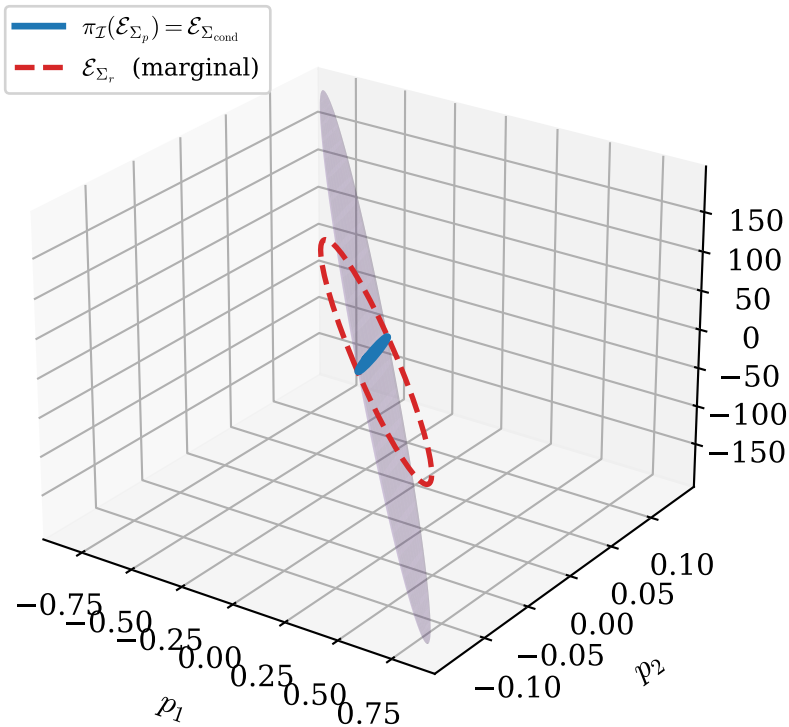


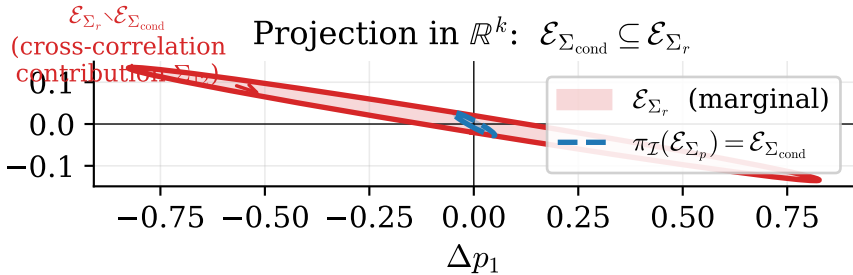
Projection of $\mathcal{E}_{\Sigma_p}$ onto retained coordinates: $\pi_{\mathcal{I}}(\mathcal{E}_{\Sigma_p}) = \mathcal{E}_{\Sigma_{\text{cond}}} \subseteq \mathcal{E}_{\Sigma_r}$

Full ellipsoid $\mathcal{E}_{\Sigma_p}$ and projection $\pi_{\mathcal{I}}$



p_3

Δp_2



Projection in $\mathbb{R}^k$: $\mathcal{E}_{\Sigma_{\text{cond}}} \subseteq \mathcal{E}_{\Sigma_r}$